The bias interval of the elephant random walk

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Abstract

Guérin, Laulin, Raschel and Simon conjectured that the initial biases giving the superdiffusive elephant random walk a bimodal limit law form an interval centered at one half. We prove this. For every superdiffusive memory, strongly biased starts are unimodal, and the limit density has two local maxima exactly when the bias lies in a proper open interval. Outside it, including its endpoints, the density has a unique local maximum. The interval is nonempty exactly above a single memory threshold, numerically about 0.6688, whose uniqueness follows from a positive-weight integral for the central curvature of the fair-start density. We control sign changes in the finite walk's first and second differences, including those created at the boundary, and pass them to the limit. The bias boundary is continuous and tends to zero as memory tends to one.

Keywords: elephant random walk; bimodality; shape transition; variation diminution; initial bias.
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1 Introduction

Fix the first step of a superdiffusive elephant random walk. Guérin, Laulin, Raschel and Simon prove that its limit density is unimodal [9, Theorem 1.1]. Randomizing the first step mixes this density with its reflection. They conjecture that the initial biases giving two peaks form an open interval centered at one half [9, Remark 2.1(b)]. We prove this and show that the interval is nonempty exactly above a unique memory threshold.

Their finite-walk induction proves conditional unimodality. For an arbitrary start, new signs can enter at the ends, so we retain the boundary values needed at later lengths in each row. This gives at most two peaks and no valley on the half favored by the first step. An affine comparison in the initial bias then gives the interval. We also prove that the conditional density has positive slope at zero for every superdiffusive memory, which makes the bimodal interval a proper subinterval of $(0, 1)$. To determine when it

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opens, we use the transform parametrization of [9, Proposition 3.2]. A Mellin calculation converts it to a positive-weight formula for the central curvature, whose single crossing gives the unique threshold.

1.1 The walk and its parameters

The elephant walk [1] takes steps $X_i \in \{+1, -1\}$. At each new step it chooses an earlier step uniformly and copies its direction, except with probability ε, when it reverses it. The choices and reversal coins are independent, also of the first step. Put

$$A_N = \#\{i \le N : X_i = +1\}, \qquad S_N = 2A_N - N, \qquad P_N(k) = \mathbb{P}(A_N = k).$$

We use $a = 1 - 2\varepsilon$ for memory and $q = \mathbb{P}(X_1 = 1)$ for the initial bias. The original copying probability is $p = (1 + a)/2$. The next-step probability at count k is $r_t(k) = \varepsilon + ak/t$. A fair start means $q = 1/2$. The notation $\mathbb{P}_+$ and $\mathbb{E}_+$ conditions on $X_1 = 1$.

For $a \in (1/2, 1)$, S_N/N^a has an almost-sure limit [6, 7]. Denote its conditional density by h_a. For initial bias q, write

$$g_{a,q}(u) = qh_a(u) + (1 - q)h_a(-u), \qquad g_a = g_{a,1/2}.$$

1.2 The bias interval and its threshold

Theorem 1.1 (the bias interval)**.** *For every $a \in (1/2, 1)$ there is $b_a \in (0, 1/2]$ such that $g_{a,q}$ has exactly two local maxima if and only if $b_a < q < 1 - b_a$. Outside this open interval, including its endpoints, the density is unimodal with a unique local maximum.*

Thus strongly biased starts are always unimodal. The next theorem determines which memories admit two peaks.

Theorem 1.2 (the analytic threshold)**.** *There is exactly one $a_* \in (2/3, 69/100)$ with $g''_{a_*}(0) = 0$. The central curvature is negative below a_* and positive above it, and $\partial_a g''_a(0)|_{a=a_*} > 0$. The bias interval is empty exactly when $a \le a_*$. The function $a \mapsto b_a$ is continuous on $(1/2, 1)$, continuously differentiable on $(a_*, 1)$, strictly decreasing immediately above a_*, and tends to zero as $a \uparrow 1$. The fair density has exactly two inflection points when unimodal and four when bimodal. At a_* it has a quartic top. For positive constants K_*, Q_*, its positive mode and bias boundary satisfy*

$$\kappa_a \sim K_* \sqrt{a - a_*}, \qquad \tfrac{1}{2} - b_a \sim Q_*(a - a_*)^{3/2} \quad (a \downarrow a_*).$$

Theorems 3.1 and 3.3 give the shape classification. Theorem 3.2 proves the unique transition, Theorem 3.5 gives the boundary regularity, and Corollary 3.6 gives the limit at one. None of these arguments uses a numerical certificate. Lemma B.2 and Propositions 3.7 and 3.8 give the critical shape and opening laws. Global decrease of b_a remains open.

Proposition 1.3 (numerical constants, computer-assisted)**.** *The threshold is $a_* \approx 0.6688$. The positive constants in Theorem 1.2 satisfy $K_* \approx 7.224$ and $Q_* \approx 9.867$.*

Propositions 3.7 and 3.8 give the expansions. Appendix C records the certified enclosures, with their error analysis in the supplement.

1.3 From finite signs to the limiting shape

The interior recurrence takes positive sums of neighbors, so it cannot create a sign change. The ends need separate treatment. We enlarge each row by the boundary values that will enter later (Figure 2). Bounding their signs first closes the induction for the whole row and for the half favored by the initial bias. Lemma A.1 passes the resulting bounds directly to the density and its derivatives.

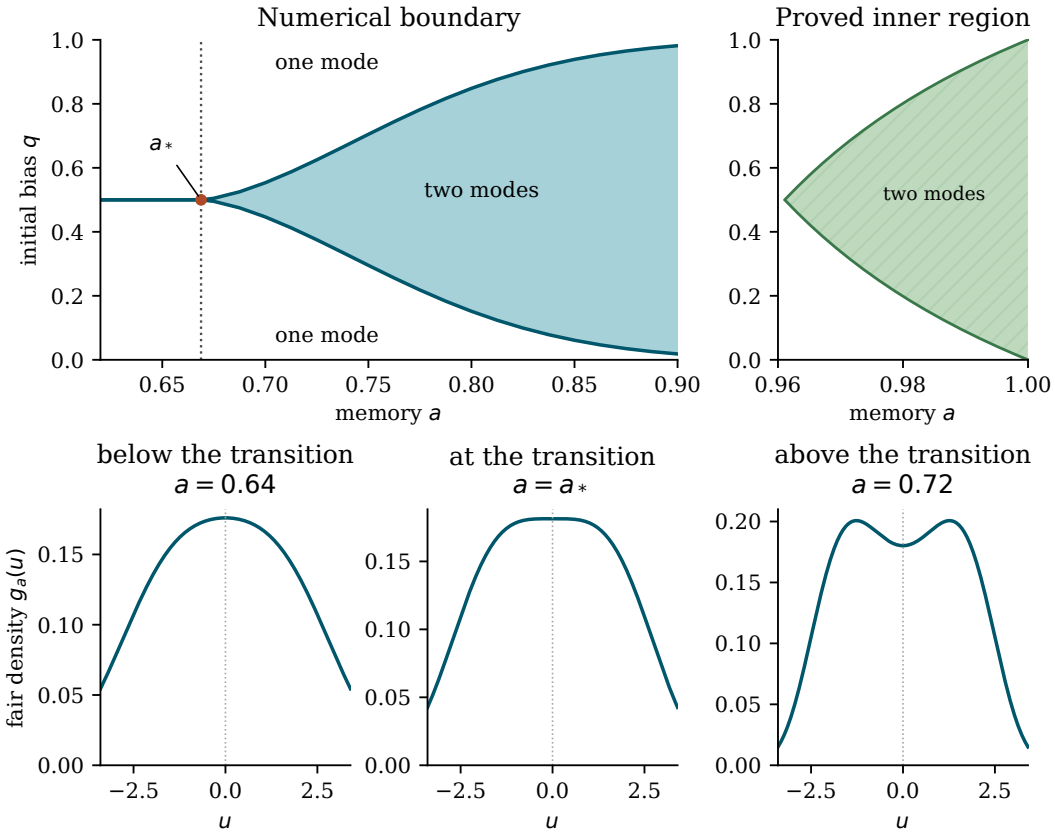


Figure 1: The bias interval and the fair-start transition. The upper left panel shows numerical values of the continuous boundary in Theorem 3.5 (the curves and shading are not certified enclosures). The upper right panel plots the explicit inner region $4\rho_a/(1-4\rho_a) < q < 1 - 4\rho_a/(1-4\rho_a)$ proved in Corollary 3.6. It does not locate the full boundary. The lower panels show the fair density before, at and after the transition. Lemma B.2 proves analytically that the top at the threshold is quartic. Appendix C gives a certified enclosure for the threshold's location.

For $q \geq 1/2$, the right half has no valley and dominates its reflection. A second peak therefore occurs exactly when the density decreases somewhere on the left. At two fixed points there, the difference is affine in q and has the opposite sign at $q = 1$. Once a decrease occurs, it persists as q moves toward one half. Taking all pairs and reflecting gives the bias interval.

For the memory threshold, an independent exponential scale gives the explicit transform of [9, Proposition 3.2]. Two integrations by parts in its Mellin transform express the central curvature as a weighted mean minus a constant. Rational bounds fix the sign outside $[2/3, 69/100]$. Within that interval, an increasing score and a covariance identity give exactly one crossing (Appendix B).

1.4 Prior work

The counting chain predates the elephant walk. The Drezner–Farnum model [2], in the formulation of James, James and Qi [4, p. 2339], has next-success probability $(1-\theta)p + \theta k/t$. Taking $p = 1/2$ and $\theta = a$ gives our transition rule. Their family includes bimodal laws. Heyde [3] established fixed-parameter limit regimes, and James, James

and Qi treat time-varying dependence. Baur and Bertoin [6] derive functional limits through the connection with Pólya-type urns. Further limit and fluctuation results include [5, 7, 8].

We use the density and moments of [10, Theorems 1.3 and 1.4] and conditional unimodality from [9, Theorem 1.1]. Kiss and Vető [11] give low-order moments for general steps. Recent preprints study transition probabilities, Gaussian rates and moment recurrences [12, 14, 15]. Trinas and Valle [13] prove a superdiffusive local limit and disprove conditional log-concavity. Their difference recurrence also yields our derivative-limit lemma, whose proof is included here, so no result depends on the unrefereed preprint.

Mohandas et al. [21, Supplemental Material, Section IV.A] observe the fair-start transition in a growing-network model. They extrapolate finite-length central curvatures to a copying probability near 0.836. Our threshold is $(1 + a_*)/2 \approx 0.8344$ in their parameter.

Which mixing weights give two peaks is a classical question. Eisenberger [24] and Robertson and Fryer [25] study it for normal mixtures. Here the components are a non-Gaussian limit density and its reflection. The finite sign argument uses classical variation diminution [23, 22], with the extra work of controlling boundary histories before applying the positive interior recurrence.

2 Finite shape bounds

Theorem 2.1 (the finite shape bounds)**.** *For every $N \geq 1$, every $0 < \varepsilon < 1$ and every initial bias, the count law has at most two modal plateaus, including its endpoints. If $q \geq 1/2$, it dominates its reflection on the right half and has no strict valley there. For a fair start, $N \geq 3$ and $0 < \varepsilon < 1/2$, the nonzero signs of*

$$D_N(k) = P_N(k+1) - 2P_N(k) + P_N(k-1), \qquad 1 \leq k < N,$$

after equal neighbors are merged, form a subsequence of $(+, -, +, -, +)$.

Propositions A.4 and A.5 prove the first-difference assertions. Theorem 2.5 proves the second-difference assertion.

The second difference of P_N changes sign at most four times under a fair start. This bounds the limiting density's inflection points, while first differences bound its modes. Appendix A gives the first-difference argument for arbitrary initial bias and the passage to the limit.

For the fair-start second differences, $0 < \varepsilon < \frac{1}{2}$ and $a = 1 - 2\varepsilon$, so that $r_t(b) = \varepsilon + ab/t$. For $t \geq 1$ and every integer k we write $P_t(k) = \mathbb{P}(A_t = k)$, which is 0 unless $0 \leq k \leq t$, and

$$D_t(k) = P_t(k+1) - 2P_t(k) + P_t(k-1).$$

We also put $s_t = D_t(1) = P_t(0) - 2P_t(1) + P_t(2)$ for $t \geq 2$.

Lemma 2.2 (recurrence for the second difference)**.** *For every $t \geq 1$ and every integer k,*

$$P_{t+1}(k) = \big(1 - r_t(k)\big)P_t(k) + r_t(k-1)\,P_t(k-1), \tag{2.1}$$

$$D_{t+1}(k) = \big(1 - r_t(k+1)\big)D_t(k) + r_t(k-2)\,D_t(k-1). \tag{2.2}$$

Also $P_t(k) = P_t(t-k)$ and $D_t(k) = D_t(t-k)$.

Proof. Given $A_t = k$, the copied step is $+1$ with probability k/t, so step $t+1$ is $+1$ with probability $(1-\varepsilon)\frac{k}{t} + \varepsilon(1 - \frac{k}{t}) = r_t(k)$. This gives (2.1) for $0 \leq k \leq t+1$. At $k = 0$ and $k = t+1$, the extra term $r_t(-1)P_t(-1)$ or $(1 - r_t(t+1))P_t(t+1)$ is 0 because $P_t(-1) = P_t(t+1) = 0$. For other k every term vanishes. Reversing every step preserves

the fair-start law, giving the stated symmetry. Taking second differences gives the symmetry of D_t.

For (2.2), take second differences in (2.1). The affine identity $r_t(k+1) - r_t(k) = a/t$ gives

$$2r_t(k+1) + r_t(k-2) = 3r_t(k), \qquad r_t(k+1) + 2r_t(k-2) = 3r_t(k-1),$$

so the coefficients of $P_t(k)$ and $P_t(k-1)$ agree with the right side of (2.2). The outer two coefficients agree directly. □

Trinas and Valle [13, (2.2)] use (2.1), the forward equation, for a superdiffusive local limit theorem. Their higher-difference recurrence [13, (4.10)] supports a certified fixed-memory study of log-concavity. We use (2.2) to count sign changes. The coefficients are positive for $2 \le k \le t-1$, so only the ends can introduce new changes. The next lemma explains why positive sums cannot create them in the interior (see [23, 22] for the general theory).

For a finite real list z, define its sign pattern $\mathrm{sp}(z)$ by deleting the zeros of z, replacing each remaining entry by its sign and merging equal neighbors. Thus $\mathrm{sp}(z)$ lists the alternating signs. For two such lists we write $p \preceq q$ if p is a subsequence of q. We put $\Pi_5 = (+,-,+,-,+)$ and $\Pi_3 = (+,-,+)$.

Lemma 2.3 (positive sums of neighbors)**.** *Let $\zeta = (\zeta_1, \dots, \zeta_n)$ be real, and let $\eta_i = A_i\zeta_i + B_i\zeta_{i+1}$ for $1 \le i \le n-1$ with $A_i \ge 0$ and $B_i \ge 0$. Let $w = (\zeta_1, \eta_1, \zeta_2, \eta_2, \dots, \eta_{n-1}, \zeta_n)$. Then $\mathrm{sp}(w) = \mathrm{sp}(\zeta)$ and $\mathrm{sp}(\eta) \preceq \mathrm{sp}(\zeta)$.*

We prove this in Appendix D.1.

For the walk, we still need the rows and their initial signs. Figure 2 shows the first-difference construction. Proposition A.4 checks the full rows and Proposition A.5 the half favored by the first step.

$$\underbrace{\ell_N, \dots, \ell_{t+1}}_{\text{left boundary history}} \qquad \underbrace{d_t(0), \dots, d_t(t-1)}_{\text{active row}} \qquad \underbrace{u_{t+1}, \dots, u_N}_{\text{right boundary history}}$$

$$\downarrow \text{ copy or sum adjacent entries}$$

$$\ell_N, \dots, \ell_{t+2} \qquad \ell_{t+1}, d_{t+1}(1), \dots, d_{t+1}(t-1), u_{t+1} \qquad u_{t+2}, \dots, u_N$$

Figure 2: Fix a final length N and retain the boundary values that will enter later. Here $d_t(k) = P_t(k+1) - P_t(k)$, $\ell_t = d_t(0)$ and $u_t = d_t(t-1)$. The active row grows by one while the augmented row shrinks by one. Each incoming end is already in the history, so its sign is included in the preceding row's bound.

Lemma 2.4 (sign changes next to an end)**.** *For every $t_1 \ge 2$ we have $\mathrm{sp}(s_2, s_3, \dots, s_{t_1}) \preceq \Pi_3$.*

We prove this in Appendix D.2.

Theorem 2.5 (at most five blocks)**.** *Let $N \ge 3$ and $0 < \varepsilon < \frac{1}{2}$. Then*

$$\mathrm{sp}\big(D_N(1), D_N(2), \dots, D_N(N-1)\big) \preceq (+,-,+,-,+).$$

So the second difference of P_N changes sign at most four times on $\{1, \dots, N-1\}$.

We prove this in Appendix D.3.

3 The limit density at fixed memory

Fix $\varepsilon \in (0, \frac{1}{4})$, so $a = 1 - 2\varepsilon \in (\frac{1}{2}, 1)$. Conditional on $X_1 = 1$, S_k/k^a converges almost surely to $\mathcal{L}_a$ [7, Theorem 3.7]. The fair-start limit $\mathcal{L}_a^\circ$ has the law of $\xi\mathcal{L}_a$, where ξ is a

fair sign independent of $\mathcal{L}_a$ [10, (1.3) and (1.6)]. Guérin, Laulin and Raschel prove that both limits have positive, bounded and smooth densities [10, Theorem 1.3], denoted by h_a and g_a as before. Guérin, Laulin, Raschel and Simon prove that $\mathcal{L}_a$ is unimodal [9, Theorem 1.1].

3.1 Modes and inflection points

A probability density h on $\mathbb{R}$ is *unimodal* if for some m it is nondecreasing on $(-\infty, m]$ and nonincreasing on $[m, \infty)$. Guérin, Laulin, Raschel and Simon ask for a unique local maximum [9], so we must also exclude a flat top. The density g_a mixes the unimodal density of $\mathcal{L}_a$ with its reflection. Such a mixture can have more than one peak [18]. We first take a fair start. Lemma A.3 excludes constant intervals, so modes are points. We consider other biases afterward.

Theorem 3.1 (at most two modes)**.** *Let $a \in (\frac{1}{2}, 1)$. Exactly one of the following holds.*

(a) The density g_a is nondecreasing on $(-\infty, 0]$ and nonincreasing on $[0, \infty)$.

(b) There is $\kappa > 0$ such that g_a is nondecreasing on $[0, \kappa]$ and nonincreasing on $[\kappa, \infty)$, and $g_a(0) < g_a(\kappa)$.

In case (b), g_a is nondecreasing on $(-\infty, -\kappa]$ and nonincreasing on $[-\kappa, 0]$, and it is not unimodal. All these monotonicities are strict. The unique local maximum is 0 in case (a), and the only local maxima are $\pm\kappa$ in case (b). In case (a), there is $d > 0$ such that g_a is strictly concave on $(-d, d)$ and strictly convex on each tail. In case (b), there are $0 < d_1 < \kappa < d_2$ such that g_a is strictly convex on $(-d_1, d_1)$ and on each tail beyond $\pm d_2$, and strictly concave on $(-d_2, -d_1)$ and (d_1, d_2). Thus it has exactly two inflection points in case (a) and four in case (b).

A real function v has *values at most* Π if $\operatorname{sp}(v(y_1), \dots, v(y_j)) \preceq \Pi$ for every $y_1 < \cdots < y_j$.

Proof. Fix $a \in (1/2, 1)$. Lemma A.1 at order two passes the five-block bound of Theorem 2.5 to g_a''. Indeed, the interpolant's values are positive sums of adjacent second differences, and its lattice fills every fixed compact interval as $N \to \infty$. Lemmas 2.3 and A.2 therefore give values at most Π_5.

Symmetry. Proposition A.5 and Lemma A.1, with $r = 0$, show that g_a has no strict valley on $[0, \infty)$. Such a valley would persist at nearby lattice points under locally uniform convergence.

If every increment on $[0, \infty)$ is nonpositive, case (a) holds. Otherwise choose $y_1 < y_2$ with $g_a(y_1) < g_a(y_2)$. Integrability supplies $y_3 < y_4$, both beyond y_2, with $g_a(y_3) > g_a(y_4)$. The no-valley property excludes any later increase, so g_a attains a maximum on $[0, \infty)$ at some κ. The absence of a strict valley makes it nondecreasing before κ and nonincreasing afterward. The positive increment forces $\kappa > 0$. If $g_a(0) = g_a(\kappa)$, it would be constant on $[0, \kappa]$, contrary to analyticity and the positive increment. Thus $\kappa > 0$ and case (b) holds. Evenness gives the left-half shape and excludes unimodality in (b).

Equality at two points of a monotone interval would force a constant interval, contrary to Lemma A.3. Thus the monotonicities are strict and there is no additional local maximum.

Curvature. The Fourier bound in Lemma A.3 gives $g_a(y), g_a'(y) \to 0$ as $y \to \infty$. Evenness and the Π_5 bound give at most three sign blocks for g_a'' on $(0, \infty)$. Its last sign must be positive. Otherwise g_a' would eventually be nonincreasing with limit zero, making g_a eventually nondecreasing. This contradicts $g_a > 0$ and $g_a \to 0$. Analyticity and strict monotonicity imply that the first nonconstant term in the even Taylor series of g_a at zero is negative in case (a) and positive in case (b). The same signs therefore hold for g_a''

near zero. In case (b), g_a'' must also be negative somewhere, since g_a' is positive near zero and negative somewhere beyond κ. The positive-half-line sign patterns are consequently $(-,+)$ in case (a) and $(+,-,+)$ in case (b). Analyticity excludes an interval of zero curvature and makes the resulting convexities strict. In case (b), $g_a'(d_1) > 0 > g_a'(d_2)$, so $d_1 < \kappa < d_2$. □

Theorem 3.2 (the unique transition)**.** *There is exactly one $a_* \in (1/2, 1)$ with $g_{a_*}''(0) = 0$. It lies in $(2/3, 69/100)$, and*

$$g_a''(0) < 0 \quad (1/2 < a < a_*), \qquad g_a''(0) > 0 \quad (a_* < a < 1).$$

Moreover, $\partial_a g_a''(0)|_{a=a_*} > 0$. The density g_a is unimodal for $a \le a_*$ and bimodal for $a > a_*$. In particular, the critical density is unimodal.

To rule out a second crossing, Appendix B expresses the curvature sign as a comparison of positive weighted means, without numerical input. On the short interval where the sign can change, an increasing score and a covariance identity make the weighted mean strictly increase with memory. Together with the endpoint signs, this gives exactly one crossing. Appendix C separately locates it near 0.6688.

3.2 Arbitrary initial bias

Write h_a for the density of $\mathcal{L}_a$ and $g_{a,q}(x) = qh_a(x) + (1-q)h_a(-x)$ for initial bias $\mathbb{P}(X_1 = 1) = q$. Thus $g_{a,1/2} = g_a$.

Theorem 3.3 (the interval of bimodal biases)**.** *For every $a \in (1/2, 1)$ and $q \in [0, 1]$, the density $g_{a,q}$ has at most two local maxima. There is $b_a \in [0, 1/2]$ such that it has exactly two local maxima if and only if*

$$b_a < q < 1 - b_a.$$

Outside this open interval, including its endpoints, the density is unimodal with a unique local maximum. The interval is empty exactly when $a \le a_*$, in which case $b_a = 1/2$. At finite time, for every $0 < \varepsilon < 1$, $q \in [0, 1]$ and $N \ge 1$, the count law has at most two modal plateaus, including its endpoints.

Proof. The mode bound. Proposition A.4 gives the finite assertion and excludes two successive strict valleys. The locally uniform density convergence in Lemma A.1, with $r = 0$, passes these inequalities to $g_{a,q}$. Lemma A.3 excludes constant intervals for every bias. Hence three local maxima $x < y < z$ would give $x < u < y < v < z$ with

$$g_{a,q}(u) < \min\{g_{a,q}(x), g_{a,q}(y)\}, \qquad g_{a,q}(v) < \min\{g_{a,q}(y), g_{a,q}(z)\}.$$

For each pair, take a one-sided neighborhood of the smaller maximum toward the other one. Since the density is not constant on an interval, some point there has strictly smaller value. This gives the two forbidden valleys.

The bias interval. Fix a, write $f_q = g_{a,q}$ and $h = h_a$, and call $x < y < z$ a strict valley if $f_q(y) < \min\{f_q(x), f_q(z)\}$. Proposition A.5 and the same density convergence show that, for $q \ge 1/2$, f_q has no strict valley on $[0, \infty)$ and $f_q(-x) \le f_q(x)$ for $x \ge 0$. Thus it has a strict valley if and only if $f_q(x) > f_q(y)$ for some $x < y \le 0$. Reflection dominance turns such a descent into the valley $(x, y, -x)$. Conversely, a valley must start at $x < 0$. If its middle point y is positive, the no-valley property at $(0, y, z)$ gives $f_q(0) \le f_q(y) < f_q(x)$.

At $q = 1$, conditional unimodality [9] and reflection dominance place the mode in $[0, \infty)$. A negative mode would have a reflected point of at least the same height, contrary to uniqueness. Thus h is nondecreasing on $(-\infty, 0]$. For fixed $x < y \le 0$, the difference $f_q(x) - f_q(y)$ is affine in q and nonpositive at $q = 1$. If positive at $q \ge 1/2$, it remains positive for every $r \in [1/2, q]$. Thus, within $[1/2, 1]$, a valley persists when the bias

decreases. These biases form a relatively open set because the strict inequalities at three fixed points survive a small change of bias. The set excludes 1, and reflection gives an open interval $(b_a, 1-b_a)$, possibly empty.

A strict valley gives two local maxima. Integrability and positivity let us choose an outer point on each side below the corresponding high point. We then maximize on the two resulting compact intervals. Conversely, two local maxima give a strict valley because there is no flat interval. The mode bound excludes a third maximum. A continuous integrable density without a strict valley is unimodal, and absence of flat intervals makes its maximum unique. Finally the bias interval is nonempty exactly when it contains $1/2$, so Theorem 3.2 identifies the empty case. □

This proves the initial-bias conjecture of [9, Remark 2.1(b)], with its half-width $q(a) = 1/2 - b_a$. The derivative-limit argument used here is proved in Appendix A.

3.3 The bias boundary

To find the endpoint, we compare the conditional density's slopes at reflected points. Their ratio tells us how much weight the reflected density needs to create a decrease.

Proposition 3.4 (a variational equation for the bias boundary)**.** *For $a \in (1/2, 1)$ put $U_a = \{t > 0 : h_a'(t) > 0\}$. With the infimum over an empty set interpreted as $+\infty$,*

$$b_a = \min\left\{\frac{1}{2}, \quad \inf_{t \in U_a} \frac{h_a'(-t)}{h_a'(t) + h_a'(-t)}\right\}. \tag{3.1}$$

In particular $b_a \to 0$ as $a \uparrow 1$. If the infimum is attained at $t > 0$ with $h_a'(t) > 0$ and $b_a < 1/2$, then, with $q = 1 - b_a$,

$$g_{a,q}'(-t) = g_{a,q}''(-t) = 0.$$

At this point the smaller peak disappears.

Proof. The proof of Theorem 3.3 shows that $h_a'(-t) \geq 0$ for $t \geq 0$. For $q \geq 1/2$ bimodality is equivalent to a decrease somewhere on the negative half-line, hence to $g_{a,q}'(-t) < 0$ for some $t > 0$. Here

$$g_{a,q}'(-t) = q h_a'(-t) - (1-q) h_a'(t).$$

This expression can be negative only if $t \in U_a$, in which case it is negative exactly when $1 - q > h_a'(-t)/(h_a'(t) + h_a'(-t))$. Taking the union over t gives (3.1), with equality excluded as in the interval theorem. At an interior minimizer the ratio has derivative zero. Differentiating its numerator and denominator gives the two displayed equations. The moment estimate in Corollary 3.6 gives $b_a \to 0$. □

Theorem 3.5 (regularity of the bias boundary)**.** *The function $a \mapsto b_a$ is continuous on $(1/2, 1)$, is continuously differentiable on $(a_*, 1)$, and extends continuously to $b_1 = 0$. It is strictly decreasing on some interval immediately above a_*. For every $a > a_*$, we have $0 < b_a < 1/2$, and the infimum in (3.1) has a unique minimizer $t_a \in (0, \kappa_a)$. This minimizer depends continuously on a throughout $(a_*, 1)$.*

Proof. For $a > a_*$, the curvature partition in Theorem 3.1 gives $g_a'(t) > 0$ on $(0, \kappa_a)$ and $g_a'(t) < 0$ on (κ_a, ∞). For $t \geq \kappa_a$ in U_a, the ratio in (3.1) is therefore at least $1/2$. Since $b_a < 1/2$, it is enough to search $(0, \kappa_a)$ (we still have to show the infimum is attained). Put

$$D_a(t) = h_a'(t) + h_a'(-t), \qquad R_a(t) = \frac{g_a'(t)}{D_a(t)} \quad (0 < t < \kappa_a).$$

Lemmas A.6 and A.7 give $h'_a(-t) > 0$ for every $t \geq 0$. We therefore have $D_a > 0$, $0 < R_a < 1/2$, and $1/2 - b_a = \sup_{0<t<\kappa_a} R_a(t)$. Also $R_a(0+) = 0$.

We first rule out a strict valley in the ratio. We claim that

$$R_a(v) \geq \min\{R_a(u), R_a(w)\} \qquad (0 < u < v < w < \kappa_a). \tag{3.2}$$

Otherwise, choose $s \in (0, 1/2)$ strictly between the middle value and both outer values. With $q = 1/2 + s$, $g'_{a,q}(-t) = D_a(t)(s - R_a(t))$ then has signs $-, +, -$ at $-w, -v, -u$, and is positive sufficiently close to zero on the left. Positivity of the density and its zero limits at infinity supply a positive derivative farther left and a negative derivative farther right. There would be three local maxima, contrary to Theorem 3.3.

At κ_a, we have $D_a(\kappa_a) = 2h'_a(-\kappa_a) > 0$. Thus R_a extends continuously to $[0, \kappa_a]$, with value zero at both endpoints. Its positive maximum is attained in the interior. At that point $h'_a(-t) > 0$, so the maximum is strictly below $1/2$. Hence $b_a > 0$. We need attainment here (pointwise strict inequality on an open interval would not suffice). Equation (3.2) and analyticity make this maximum unique, since two maximizers would force a constant interval. They also make R_a strictly increasing before its maximizer and strictly decreasing after it.

Lemma A.3 gives joint continuity of all the derivatives used here. The signs of g'_a on either side of its unique positive zero give continuity of κ_a. Fix $a_0 > a_*$ and choose $0 < l < t_{a_0} < r < \kappa_{a_0}$. The strict inequalities $R_a(l), R_a(r) < R_a(t_{a_0})$ persist near a_0. For $v < l$, apply (3.2) to v, l, t_{a_0}. Since $R_a(l) < R_a(t_{a_0})$, it gives $R_a(v) \leq R_a(l)$. Applying it to t_{a_0}, r, v gives $R_a(v) \leq R_a(r)$ for $v > r$. Thus every nearby global maximum lies in the fixed interval $[l, r]$. The ratios converge uniformly on this fixed compact interval, so their maxima converge. Uniqueness gives convergence of the maximizers too. This proves continuity above a_*.

At a_*, the curvature partition in Theorem 3.1(a) makes g'_{a_*} strictly decrease from zero and then strictly increase to zero. Thus $g'_{a_*}(r) < 0$ for every $r > 0$. Joint continuity gives $g'_a(r) < 0$ for all $a > a_*$ sufficiently close to a_*, so $\kappa_a < r$. Since r was arbitrary, $\kappa_a \to 0$. On the shrinking interval $[0, \kappa_a]$, joint continuity now gives

$$\sup_{0\leq t\leq\kappa_a} |g'_a(t)| \longrightarrow 0, \qquad \inf_{0\leq t\leq\kappa_a} D_a(t) \longrightarrow 2h'_{a_*}(0) > 0,$$

where the strict inequality is Lemma A.7. Consequently $1/2 - b_a = \max R_a \to 0$. Below the threshold $b_a = 1/2$, so this proves continuity at a_* without the critical expansion. Proposition 3.4 gives the limit at one.

For differentiability, the same three-point argument puts all nearby maximizers in one compact interval. If $M(a) = 1/2 - b_a$, maximality gives, for $s > 0$,

$$R_{a+s}(t_a) - R_a(t_a) \leq M(a+s) - M(a) \leq R_{a+s}(t_{a+s}) - R_a(t_{a+s}).$$

Divide by s and use the continuity of t_a and $\partial_a R_a$ from Lemma A.3. The argument for $s < 0$ is the same. Thus

$$-b'_a = \frac{\partial_a h'_a(t_a)\, h'_a(-t_a) - h'_a(t_a)\, \partial_a h'_a(-t_a)}{\{h'_a(t_a) + h'_a(-t_a)\}^2}. \tag{3.3}$$

The right side is continuous. No derivative of t_a is needed.

Finally, put $C = \partial_a g''_a(0)|_{a=a_*} > 0$ and $H = h'_{a_*}(0) > 0$, by Theorem 3.2 and Lemma A.7. Extend $R_a = g'_a/D_a$ to a fixed neighborhood of zero, where $D_a > 0$ for a close to a_*. Evenness and Lemma A.3 give, jointly as $(a, t) \to (a_*, 0)$,

$$g'_a(t) = o(t), \qquad \partial_a g'_a(t) = (C + o(1))t, \qquad D_a(t) = 2H + o(1), \qquad \partial_a D_a(t) = O(1).$$

For the first two relations, integrate g_a'' and $\partial_a g_a''$ from zero to t, using $g_{a_*}''(0) = 0$. It follows that

$$\partial_a R_a(t) = \left(\frac{C}{2H} + o(1)\right) t > 0$$

for all sufficiently small $t > 0$ and a sufficiently close to a_*. Since $0 < t_a < \kappa_a \to 0$, the envelope formula gives $b_a' < 0$ on some interval immediately above the threshold. □

Corollary 3.6 (a rate for the bias boundary near one)**.** *Put*

$$\rho_a = \frac{\Gamma(1+a)^2}{(2a-1)\Gamma(2a)} - 1.$$

If $\rho_a < 1/12$*, then*

$$0 \le b_a \le \frac{4\rho_a}{1-4\rho_a}.$$

In particular, as $a \uparrow 1$*,* $b_a \le 8(1-a) + O((1-a)^2)$*.*

Proof. The conditional limit has mean $\mu_a = 1/\Gamma(1+a) > 0$ and variance $\rho_a \mu_a^2$, by the moment formulas of [10, Theorem 1.4]. Consider the three adjacent intervals of length μ_a centered at $-\mu_a$, 0 and μ_a. Chebyshev's inequality puts at least $1 - 4\rho_a$ of the conditional law in the right interval and at most $4\rho_a$ in the middle. For a start to the left, reflect these intervals. After mixing, the left interval has mass at least $(1-q)(1-4\rho_a)$ and the right at least $q(1-4\rho_a)$. The middle still has mass at most $4\rho_a$.

For $4\rho_a/(1-4\rho_a) < q < 1 - 4\rho_a/(1-4\rho_a)$, the middle has less mass than either outer interval. All three intervals have the same length. So, by continuity, we can choose a point in each with the middle value strictly smaller than the other two. This rules out unimodality. The two-mode bound in Theorem 3.3 gives exactly two modes, and therefore the stated bound on b_a. Finally, the logarithmic derivative at $a = 1$ of $1 + \rho_a$ is $2\psi(2) - 2 - 2\psi(2) = -2$. Thus $\rho_a = 2(1-a) + O((1-a)^2)$, which gives the linear bound. □

3.4 The density at the threshold

How far apart are the two modes just after the transition? We compute this from derivatives at the threshold. Write

$$C_* = \partial_a g_a''(0)|_{a=a_*}, \qquad B_* = -g_{a_*}^{(4)}(0), \qquad H_* = h_{a_*}'(0).$$

These constants are positive by Theorem 3.2 and Lemmas B.2 and A.7. Table 1 gives their numerical enclosures. The coefficient Q_* is defined in Proposition 3.8.

Proposition 3.7 (the critical shape)**.** *The density at the threshold satisfies*

$$g_{a_*}(u) = g_{a_*}(0) - \frac{B_*}{24} u^4 + O(u^6) \qquad (u \to 0).$$

For $a > a_*$ *let* $\kappa_a > 0$ *be the positive mode of* g_a*. As* $a \downarrow a_*$*,*

$$\kappa_a = \sqrt{\frac{6C_*}{B_*}}\sqrt{a - a_*} + o\big(\sqrt{a-a_*}\big),$$
$$g_a(\kappa_a) - g_a(0) = \frac{3C_*^2}{2B_*}(a-a_*)^2 + o\big((a-a_*)^2\big).$$

Proof. Lemma A.3 gives joint continuity of all spatial derivatives and their first memory derivatives. Taylor remainders through order four are therefore uniform on a neighborhood of a_*. At criticality, evenness, $g_{a_*}''(0) = 0$ and the analytic sign $g_{a_*}^{(4)}(0) = -B_* < 0$ from Lemma B.2 give the first assertion.

Put $\delta = a - a_* > 0$. Uniform Taylor expansion gives

$$\frac{g_a'(u)}{u} = C_*\delta + o(\delta) - \frac{B_* + o(1)}{6}u^2 + O(u^4).$$

For $u = k\sqrt{\delta}$ this has the sign of $C_* - B_*k^2/6$ when k is fixed away from $\sqrt{6C_*/B_*}$. The unique positive mode lies between any two such choices with opposite signs, by Theorem 3.1. Squeezing gives the mode expansion. Substitute it into

$$g_a(u) - g_a(0) = \tfrac{1}{2}\big(C_*\delta + o(\delta)\big)u^2 - \tfrac{1}{24}\big(B_* + o(1)\big)u^4 + O(u^6)$$

to obtain the height expansion. □

Proposition 3.8 (how the bias interval opens)**.** *With C_*, B_*, H_* as defined above, put*

$$Q_* = \frac{\sqrt{2}\,C_*^{3/2}}{3H_*\sqrt{B_*}}, \qquad Q_* \approx 9.867.$$

As $a \downarrow a_*$,

$$\frac{1}{2} - b_a = Q_*(a - a_*)^{3/2} + o\big((a - a_*)^{3/2}\big). \tag{3.4}$$

The unique minimizer t_a of Theorem 3.5 satisfies

$$t_a = \sqrt{\frac{2C_*}{B_*}}\sqrt{a - a_*} + o\big(\sqrt{a - a_*}\big) = \frac{\kappa_a}{\sqrt{3}} + o\big(\sqrt{a - a_*}\big).$$

At $q = 1 - b_a$, the stationary point is $-t_a$.

Proof. Lemma A.3 supplies the required joint regularity. Theorem 3.2 and Lemma A.7 give $C_* > 0$ and $H_* > 0$.

Write $\delta = a - a_*$. By (3.1),

$$\frac{1}{2} - b_a = \max\left\{0, \sup_{t \in U_a} \frac{g_a'(t)}{h_a'(t) + h_a'(-t)}\right\}.$$

The fair-shape theorem makes the numerator nonpositive for $t \geq \kappa_a$. Proposition 3.7 gives $\kappa_a = O(\sqrt{\delta})$, and $H_* > 0$ makes both conditional derivatives positive on this shrinking interval. Uniform Taylor expansion gives, for bounded $v \geq 0$,

$$\frac{g_a'(v\sqrt{\delta})}{h_a'(v\sqrt{\delta}) + h_a'(-v\sqrt{\delta})} = \frac{\delta^{3/2}}{2H_*}\left(C_*v - \frac{B_*v^3}{6}\right) + o(\delta^{3/2}).$$

The cubic is largest at the unique positive point $v = \sqrt{2C_*/B_*}$. Maximizing on $[0, \kappa_a/\sqrt{\delta}]$, whose right endpoint tends to $\sqrt{6C_*/B_*}$, proves (3.4). Since the ratio vanishes at both endpoints, its positive maximum is attained inside $(0, \kappa_a)$. The convergence is uniform and the limiting cubic has a unique maximum, so every scaled maximizer tends to $\sqrt{2C_*/B_*}$. This proves the assertion about t_a.

□

The two powers come from the width and height of the same cubic. Near zero the derivative is approximately $C_*(a - a_*)t - B_*t^3/6$. Its positive part has width of order $\sqrt{a - a_*}$ and height of order $(a - a_*)^{3/2}$. The width sets the distance between the modes. The height sets how much bias removes the smaller peak.

4 Computer assistance

Only the numerical location and critical derivative enclosures in Appendix C use interval arithmetic. They use Arb [17] through python-flint 0.8.0 at 1600 bits, with independent endpoint checks at 2000 bits and refined constants at 2600 bits through python-flint 0.9.0. The supplement proves the truncation and tail bounds. The tagged records give the inputs, acceptance tests and rerun commands. Lean independently checks the finite sign arguments and analytic threshold proof from the classical weak limit, as detailed in the coverage map, and is not needed to referee the written proofs.

5 Open problems

The full bias boundary. Is b_a strictly decreasing on $(a_*, 1)$? It decreases near a_* (Theorem 3.5), and Figure 1 suggests the same elsewhere. By (3.3), this reduces to the sign at the unique optimizer. We do not know that sign for all memories. The moment estimate gives $b_a \le 8(1-a) + O((1-a)^2)$. What is the exact decay near one?

A finite-length threshold. Does the central curvature have exactly one zero in $(0, 1)$ at every even length? Numerical zeros near 0.6563 at $N = 4000$ and 0.6592 at $N = 8000$ suggest a slow increase toward a_*. Our limiting bias-interval theorem does not settle this finite-length question.

The Drezner–Farnum model and higher dimensions. Changing the Drezner–Farnum innovation probability changes the transition rule, so the reflected-mixture argument does not settle its shapes when $p \neq 1/2$ [2]. In higher dimensions the fixed-point theory gives densities [10], but our sign argument requires points ordered along a line.

Acknowledgments. I thank Peter Kagey, whose Problem 131 of the Open Problem Collection [16] led to the crossing question, and Aidan Deshong for helpful conversations.

A note on the use of AI

I wrote the original draft of this paper and its mathematics. I used Claude Opus 5.5 (Anthropic, through Claude Code) for literature searches, citation checks and referee-style reviews. The reviews informed the framing and abstract and suggested proving that the bias boundary is positive, that the transition is unique, and that the critical top is quartic without computation. I read every source cited in the paper except Eisenberger [24] and Robertson and Fryer [25], for which I relied on the publisher abstracts. I proposed and drafted the following additions: Lemma A.3; the strictness and inflection-point clauses of Theorem 3.1; the fourth-moment extension of Theorem S3.1 to $a \ge 4/5$; the Jensen lower bound in Theorem 4.5(a) of the archived combined version; and the weaker weak-convergence hypothesis in its Theorem 5.9(c). I used GPT-6 Astra (OpenAI, through Codex) to revise the manuscript and to revise the proofs of these additions. I also used it to develop and draft proofs, including the analytic quartic-sign argument, the memory-regularity argument, and continuity and local decrease of the bias boundary. The associated Lean files in kagey-problems/problem131/lean/PaperD were generated and extended with Claude Opus 5.5 and GPT-6 Astra through Codex. I also used Codex to run the Lean build and axiom audit recorded in the verification archive. Section 4 and the coverage map distinguish the formal proofs, their hypotheses, the written proofs and the separate numerical checks. I take responsibility for the claims, proofs and computations in the paper.

Code and data

The tagged repository contains the source, certificate records, scripts and checksums. The endpoint signs and refined constants are recorded in data/critical_shape_check.json and data/critical_constants_certificate.json. The supplement gives the complete certificate error analysis and an independent fourth-moment check. The archived combined version retains the rare-reversal results. Section 4 and the linked coverage map specify the formalization's scope.

A Derivative limits and regularity

A.1 Derivative limits

We first pass finite differences to density derivatives. Trinas and Valle [13, Proposition 2.1 and Theorem 3.1] prove the corresponding bounds and local limit at orders zero and one. The following argument derives the recurrence and treats every fixed order, also after a prescribed finite history.

Lemma A.1. *Fix $a \in (1/2, 1)$ and a finite initial history. Extend P_n by zero outside $\{0, \dots, n\}$. Suppose the rescaled count law converges weakly to a law with smooth density f. For each fixed integer $r \geq 0$, linearly interpolate*

$$\frac{n^{(r+1)a}}{2^{r+1}} \nabla^r P_n(k) \quad \text{at} \quad (2k - n - r)n^{-a}, \qquad \nabla P_n(k) = P_n(k) - P_n(k-1).$$

These interpolants converge locally uniformly to $f^{(r)}$.

Proof. Repeated subtraction in (2.1) gives

$$\nabla^r P_{n+1}(k) = \left(\frac{1+a}{2} - \frac{ak}{n}\right) \nabla^r P_n(k) + \left(\frac{1-a}{2} + \frac{a(k-r-1)}{n}\right) \nabla^r P_n(k-1).$$

For sufficiently large n, both coefficients are nonnegative for $1 \leq k \leq n + r$, with sum $1 - (r+1)a/n$. At $k = 0$ and $k = n + r + 1$, only one summand can be nonzero and its coefficient is $(1 + a)/2$, no larger than that sum once n is large. Iterating these row-sum bounds gives

$$\sup_k |\nabla^r P_n(k)| \leq C_{a,r} n^{-(r+1)a}.$$

The bounds at orders r and $r + 1$ give uniform bounds and Lipschitz constants for the interpolants. Every subsequence therefore has a further subsequence converging locally uniformly. To identify its limit, sum by parts against a smooth compactly supported function. Weak convergence then gives the distributional derivative of order r of the limiting law. The lattice spacing is $2n^{-a}$, so the displayed normalization is exactly its power $-(r + 1)$. Since f is smooth, this distributional derivative is $f^{(r)}$. Every subsequence has the same limit, proving the assertion. □

A.2 Sign patterns

We prove the sign and regularity lemmas used in Theorem 3.1, with the notation of Sections 2 and 3.1. Deleting entries can only remove signs from a list's pattern. An alternating list p satisfies $p \preceq q$ for every alternating list q with the same first sign and at least as many entries.

Lemma A.2 (pointwise limits)**.** *Let v_n and v be real functions on $\mathbb{R}$ with $v_n(y) \to v(y)$ for every y, and let Π be an alternating list. If every v_n has values at most Π, then so has v.*

Proof. Let $y_1 < \cdots < y_j$, and let I be the set of indices i with $v(y_i) \neq 0$. For all large n, $v_n(y_i)$ has the sign of $v(y_i)$ for every $i \in I$. Then $\operatorname{sp}(v(y_1), \dots, v(y_j))$ is the pattern of the list $(v_n(y_i))_{i \in I}$, which is a subsequence of $\operatorname{sp}(v_n(y_1), \dots, v_n(y_j)) \preceq \Pi$. □

Lemma A.3 (entire densities and parameter continuity)**.** *For every $a \in (\frac{1}{2}, 1)$ and $1 < r < 1/a$, the characteristic function ϕ_a of $\mathcal{L}_a$ satisfies*

$$|\phi_a(t)| \le 2e^{-c|t|^r} \qquad (t \in \mathbb{R})$$

for some $c = c(a, r) > 0$. The same c works on a sufficiently small neighborhood of any fixed a. For every compact $J \subset (1/2, 1)$ and every $\beta > 0$, there is $C_{J,\beta} < \infty$ such that

$$|\partial_t \phi_a(t)| + |\partial_a \phi_a(t)| \le C_{J,\beta}(1 + |t|)^{-\beta} \qquad (a \in J,\ t \in \mathbb{R}).$$

For every integer $k \ge 0$, the derivatives $\partial_u^k h_a(u)$ and $\partial_a \partial_u^k h_a(u)$ exist and are jointly continuous in (a, u). For every $q \in [0, 1]$, the density of the limit with first-step bias q extends to an entire function and is not constant on an interval of positive length.

Proof. We first check weak continuity in a to obtain a Fourier bound local in memory. On a compact $J \subset (1/2, 1)$, the martingale $\mathcal{M}_n(a) = \Gamma(n)S_n/\Gamma(n+a)$ of [7, (2.4), Appendix A] satisfies

$$\sup_{a \in J} \mathbb{E}|\mathcal{L}_a - \mathcal{M}_n(a)|^2 \le C_J \sum_{k \ge n} k^{-2 \min J} \longrightarrow 0.$$

The walk's centered next step has conditional variance at most one, and the gamma ratios are uniformly bounded on J. The finite laws are continuous in a, so $\phi_a(t)$ is jointly continuous, uniformly on compact frequency intervals.

The fixed-point equation of [10, Theorem 1.1 and (2.10)] gives

$$|\phi_a(t)| \le \int_0^1 |\phi_a(tv^a)|\, |\phi_a(t(1-v)^a)|\, dv.$$

Here ϕ_a is the characteristic function conditional on a positive first step. The random sign disappears after taking absolute values, since $|\phi_a(-s)| = |\phi_a(s)|$. Fix a_0 and $1 < r < 1/a_0$, and choose J about a_0 with $r \max J < 1$. Put $D(v) = v^{r \max J} + (1-v)^{r \max J} - 1$. We have $D(v) > 0$ for $0 < v < 1$. Dominated convergence therefore permits $K > 1$ such that

$$R := 2 \int_0^1 e^{-K^r (\log 2) D(v)}\, dv < 1.$$

The law of $\mathcal{L}_{a_0}$ has a density [10, Theorem 1.3], so the Riemann–Lebesgue lemma gives $T > 0$ with $|\phi_{a_0}(t)| < 2^{-K^r}$ for $|t| \ge T$. Joint continuity, after shrinking J, gives $|\phi_a(t)| \le 2^{1-K^r}$ for $a \in J$ and $T \le |t| \le KT$. Set $c = (\log 2)/T^r$ and $b(t) = 2e^{-ct^r}$ for $t \ge 0$. On $[0, T]$ we have $b \ge 1$, and on $[T, KT]$ we have $b \ge 2^{1-K^r}$. Thus $|\phi_a| \le b$ on $[0, KT]$.

For $t \ge KT$,

$$\frac{\int_0^1 b(tv^a) b(t(1-v)^a)\, dv}{b(t)} \le 2 \int_0^1 e^{-ct^r D(v)}\, dv \le R < 1.$$

Suppose the bound $|\phi_a| \le b$ stops at a finite $t_* \ge KT$. By continuity it still holds on $[0, t_*]$. At the endpoint the fixed-point inequality gives the strict bound $|\phi_a(t_*)| \le Rb(t_*) < b(t_*)$. Continuity then extends the interval past t_*, a contradiction. The bound thus holds on $[0, \infty)$, and evenness extends it to $\mathbb{R}$.

Since $r > 1$, the Fourier integral converges absolutely and locally uniformly for complex arguments. Indeed, for every $B < \infty$, $\int_{\mathbb{R}} e^{B|t|} |\phi_a(t)|\, dt < \infty$. It therefore defines an entire function. After any fixed number of spatial derivatives, the bound on J still supplies an integrable majorant, giving joint continuity by dominated convergence. For initial bias q, the triangle inequality applied to $q\phi_a(t) + (1-q)\phi_a(-t)$ gives the same bound. A density of integral one cannot be constant on $\mathbb{R}$. If it were constant on an interval, the identity theorem would make it constant everywhere.

To differentiate in memory, we first work on bounded frequency intervals. The moment recursion of [10, Theorem 1.4] is

$$m_0 = m_1 = 1, \qquad m_n(z) = \frac{\sum_{j=1}^{n-1} \theta_j(z) m_j(z) m_{n-j}(z)}{nz - \theta_n(z)}, \qquad \theta_j(z) = \begin{cases} 1 & j \text{ even}, \\ z & j \text{ odd}. \end{cases}$$

Choose a closed complex disc about a_0 on which $\operatorname{Re} z \geq \alpha > 1/2$, $|z| < 1$ and $|\operatorname{Im} z| \leq \rho$. All denominators for $n \geq 2$ have real part at least $n\alpha - 1 > 0$. Put $A = 2\alpha - 1$ and $B = A^{-1}$. Induction gives

$$|m_n(z)| \leq AB^n \quad (n \geq 1), \qquad \frac{(n-1)A^2}{n\alpha - 1} \leq A \quad (n \geq 2).$$

For $x > 0$ and real y, the gamma product gives

$$\left| \frac{\Gamma(x)}{\Gamma(x+iy)} \right|^2 = \prod_{j \geq 0} \left(1 + \frac{y^2}{(x+j)^2} \right) \leq \exp\left(\frac{y^2}{x^2} + \frac{y^2}{x} \right).$$

Taking $x = 1 + n \operatorname{Re} z$ and $y = n \operatorname{Im} z$ gives

$$\left| \frac{m_n(z)}{\Gamma(1+nz)} \right| \leq \frac{CB_0^n}{\Gamma(1+n\alpha)}.$$

Here C, B_0 are independent of n, z. We used monotonicity of Γ on $[2, \infty)$ for $n \geq 2$, and enlarged C for $n = 0, 1$. Thus $\sum_{n \geq 0} m_n(z)(it)^n / \Gamma(1+nz)$ converges locally uniformly in complex t and z in the disc's interior. For real $z = a$ this is $\phi_a(t)$, by the moment formula and exponential moments of [10, Corollary 2.11]. Cauchy's formula gives local joint continuity of every frequency derivative and its memory derivative.

We now obtain bounds at large frequencies without a numerical input. Shrink J about a_0 so that it lies inside this disc and retains the uniform exponential bound. For $t \geq 0$ write $f_a(t) = |\phi_a(t)|$, $u_a(t) = |\partial_t \phi_a(t)|$ and $w_a(t) = |\partial_a \phi_a(t)|$. The smoothing equation is

$$\phi_a(t) = \int_0^1 \phi_a(tv^a) \{ p \phi_a(t(1-v)^a) + (1-p) \phi_a(-t(1-v)^a) \} \, dv, \qquad p = (1+a)/2.$$

The local bounds just proved justify differentiation under this integral, since $v^{\min J} |\log v|$ is integrable. Put $x = tv^a$, $y = t(1-v)^a$ and, for nonnegative z, define

$$(\mathcal{K}_a z)(t) = \int_0^1 \{ z(x) f_a(y) + f_a(x) z(y) \} \, dv.$$

Taking absolute values and discarding factors $v^a, (1-v)^a \leq 1$ in the frequency derivative gives

$$u_a \leq \mathcal{K}_a u_a, \qquad w_a \leq \mathcal{K}_a w_a + S_a,$$

where

$$S_a(t) = \int_0^1 \{ x |\log v| u_a(x) f_a(y) + y |\log(1-v)| f_a(x) u_a(y) + f_a(x) f_a(y) \} \, dv.$$

The last term bounds the derivative of p, since $p' = 1/2$ and $|\phi_a(y) - \phi_a(-y)| \leq 2 f_a(y)$.

Fix $\beta > 0$. Splitting at $1/2$ and reflecting the second half gives

$$K_{a,\beta}(t) = 2 \int_0^{1/2} \left\{ \left(\frac{1+t}{1+x} \right)^\beta f_a(y) + \left(\frac{1+t}{1+y} \right)^\beta f_a(x) \right\} dv \longrightarrow 0$$

uniformly for $a \in J$ as $t \to \infty$. Indeed, $y \geq t/2$ bounds the first term by $2(1+t)^\beta e^{-c(t/2)^r}$ and the second ratio by 2^β. Also $x \geq tv^{\max J}$, so the exponential bound and dominated convergence make the second integral tend to zero uniformly. Choose T so that $K_{a,\beta}(t) \leq 1/2$ for all $a \in J$, $t \geq T$. For $X \geq T$ let

$$U_X = \max_{a \in J,\, 0 \leq t \leq X} (1+t)^\beta u_a(t).$$

All arguments x, y are at most t. The first smoothing inequality gives

$$U_X \leq \max\{U_T, U_X/2\}, \qquad \text{hence } U_X \leq U_T.$$

The bound is independent of X. This proves arbitrary polynomial decay for u_a.

The same decay holds for S_a. On $0 < v \leq 1/2$, the first logarithmic term uses the exponential decay of $f_a(y)$ and boundedness of $u_a(x)$, with $x|\log v| \leq tv^{\min J}|\log v|$. The second uses arbitrary polynomial decay of $u_a(y)$, $f_a(x) \leq 1$ and $|\log(1-v)| \leq \log 2$. The last uses the exponential decay of $f_a(y)$. Reflection covers the other half. Thus $S_\beta := \sup_{a \in J, t \geq 0} (1+t)^\beta S_a(t) < \infty$. Defining W_X as above with w_a in place of u_a, we obtain

$$W_X \leq \max\{W_T, W_X/2 + S_\beta\} \quad \Longrightarrow \quad W_X \leq \max\{W_T, 2S_\beta\}.$$

Conjugate symmetry extends both derivative bounds to negative t. A finite cover gives the stated bounds on every compact memory interval. Since β is arbitrary, multiplication by any fixed power of t leaves an integrable uniform majorant. Differentiation under Fourier inversion and dominated convergence prove the claimed joint continuity of $\partial_a \partial_u^k h_a$ for every k. □

A.3 First differences for a biased start

Proposition A.4 (two modes for every initial bias)**.** *Let $0 < \varepsilon < 1$, let $q \in [0,1]$ be the probability that the first step is positive, and let $P_n(k)$ be the law of the number of positive steps after $n \geq 1$ steps. Extend P_n by zero outside $\{0, \dots, n\}$. The first differences $d_n(k) = P_n(k+1) - P_n(k)$ satisfy*

$$\operatorname{sp}\big(d_n(-1), d_n(0), \dots, d_n(n)\big) \preceq (+,-,+,-).$$

In particular, P_n has at most two modal plateaus, including endpoint modes. Moreover, there are no integers $i_0 < \cdots < i_4$ for which

$$P_n(i_0) > P_n(i_1) < P_n(i_2) > P_n(i_3) < P_n(i_4).$$

Proof. Put $a = 1 - 2\varepsilon$ and $r_t(k) = \varepsilon + ak/t$. The forward equation is

$$P_{t+1}(k) = (1 - r_t(k))P_t(k) + r_t(k-1)P_t(k-1), \qquad P_1 = (1-q, q).$$

Subtracting adjacent equations gives

$$d_{t+1}(k) = r_t(k-1)d_t(k-1) + (1 - r_t(k+1))d_t(k), \qquad 1 \leq k \leq t-1. \tag{A.1}$$

Both coefficients are at least $\min(\varepsilon, 1-\varepsilon) > 0$, since $r_t(j)$ is a convex combination of ε and $1-\varepsilon$ for $0 \leq j \leq t$. It remains to check the two ends.

Suppose first that $q < 1$. Since $P_t(0) = (1-q)(1-\varepsilon)^{t-1} > 0$, define

$$\xi_t = \frac{P_t(1)}{P_t(0)}, \qquad b = \frac{1-2\varepsilon}{1-\varepsilon} < 1, \qquad A = \frac{\varepsilon}{1-\varepsilon} > 0.$$

The boundary equation gives

$$\xi_{t+1} = (1 - b/t)\xi_t + A, \qquad \xi_1 = \frac{q}{1-q}.$$

Thus $\delta_t = \xi_{t+1} - \xi_t$ obeys

$$\delta_{t+1} = \frac{t-b}{t+1}\delta_t + \frac{A}{t+1}. \tag{A.2}$$

Once an increment is nonnegative, every later increment is positive. Thus ξ_t first decreases and then increases (either part may be empty). For $\ell_t = d_t(0) = P_t(0)(\xi_t - 1)$ this proves

$$\operatorname{sp}(\ell_1, \dots, \ell_N) \preceq (+, -, +).$$

If $q \le 1/2$, then $\xi_1 \le 1$ and the sharper bound is $\operatorname{sp}(\ell_1, \dots, \ell_N) \preceq (-, +)$. In particular, a positive ℓ_i cannot be followed by a negative ℓ_j. Zeros do not affect either conclusion.

Write $u_t = d_t(t-1)$. Reflection of the walk gives $u_t(q) = -\ell_t(1-q)$ and $d_t^{(1-q)}(t-1-k) = -d_t^{(q)}(k)$. For $0 < q \le 1/2$, the right boundary history therefore has pattern contained in $(-, +, -)$. For $q = 0$ it is nonpositive throughout. The conditional count is at most $t-1$, so $u_t = -P_t(t-1) \le 0$.

The interior recurrence does not control incoming boundary values. We retain them in an augmented row. Fix N and, for $1 \le t \le N$, form

$$Z_t = (\ell_N, \dots, \ell_{t+1}, d_t(0), \dots, d_t(t-1), u_{t+1}, \dots, u_N).$$

Reading the left history backward and the right history forward places each incoming endpoint in the preceding augmented row. The interior recurrence can then control the whole row. For $q \le 1/2$, its first row is contained in $(+, -, +, -)$. We must also allow zeros at $\ell_1 = u_1 = 2q - 1$. Suppose Z_1 contained four ordered signs $-, +, -, +$. If the second lay in the reversed left history, the first two would give a positive left edge followed in time by a negative one, which is impossible. Otherwise the last three lie in the right history and have signs $+, -, +$. For $q > 0$, their reflected negatives give the forbidden pattern $(-, +, -)$ in the left history at bias $1 - q$. For $q = 0$, they contradict the nonpositive right history. This also covers $N = 1$.

Every entry of Z_{t+1} is a nonnegative combination of adjacent entries of Z_t. The interior entries use (A.1). The two new boundary entries and the remaining future edge values are copied, with coefficients $(1, 0)$ or $(0, 1)$. To check the ordering, number the augmented entries from left to right. Output entry i uses input entries i and $i+1$, with a zero coefficient when copied. A nonzero output has a parent of the same sign. Consecutive opposite signs have distinct parents, since one entry cannot carry both signs. Their parent indices therefore strictly increase. Choosing parents this way for any alternating output subsequence gives the same signs in order in the input. An update therefore cannot add an alternating pattern, so $\operatorname{sp}(Z_N) \preceq (+, -, +, -)$. Reflection and sign reversal preserve this pattern, proving the claim also for $q \ge 1/2$, including $q = 1$.

Finally, extension by zero adds $d_N(-1) = P_N(0) \ge 0$ on the left and $d_N(N) = -P_N(N) \le 0$ on the right. Neither enlarges the pattern. Every modal plateau contributes a positive-to-negative change, so there are at most two. If the statement's last inequalities held, telescoping on each of their four intervals would give an adjacent difference of the same strict sign. These four differences would have signs $-, +, -, +$, a contradiction. □

Proposition A.5 (the dominant half)**.** *Under the hypotheses of Proposition A.4, suppose $q \ge 1/2$. Then $P_n(n-k) \le P_n(k)$ for $2k \ge n$, and the right half has no strict valley. That is,*

$$P_n(j) \ge \min\{P_n(i), P_n(k)\} \qquad \text{if } i < j < k \text{ and } 2i + 1 \ge n.$$

Equivalently, its first differences contain no negative value followed by a positive one. Reflection gives the corresponding left-half result when $q \le 1/2$.

Proof. Write the bias explicitly as $P_{n,q}$ and put

$$A_n(k) = P_{n,q}(k) - P_{n,1-q}(k) = P_{n,q}(k) - P_{n,q}(n-k).$$

The forward equation gives $A_{n+1}(k) = (1 - r_n(k))A_n(k) + r_n(k-1)A_n(k-1)$. At $n = 1$, $A_1(1) = 2q - 1 \geq 0$. Suppose the assertion holds at length n. At a point strictly above the new center, both parents lie in the preceding upper half and both coefficients are nonnegative. At an integer center $A_{n+1}(k) = 0$ by reflection. Terms outside the support vanish. This proves the induction. In particular, the central first difference of an odd row is nonnegative.

We now keep only the half favored by the start, namely the spatial indices $2k + 1 \geq t$ in Z_t. At $t = 1$ its pattern is contained in $(+, -)$. The central entry is $2q - 1 \geq 0$, followed by the future right boundary values $u_s(q) = -\ell_s(1 - q)$. Since $1 - q \leq 1/2$, their pattern is $(+, -)$ by (A.2). We now rule out a negative difference followed by a positive one, also among the future boundary entries. Suppose such a pair first appeared in the next row. Choose a same-sign parent for each entry. The preceding proof shows that the first parent comes before the second. The sole possible parent outside the previous half occurs at the new central difference of an odd row. That entry is nonnegative, so it cannot begin the forbidden pair. More explicitly, if a negative entry at index i of row $t + 1$ satisfies $2i + 1 \geq t + 1$, then $2i > t$, so both its parents satisfy $2k + 1 \geq t$. Parents of a later positive entry follow them strictly, as in the preceding proof. Induction therefore excludes the pair in every restricted augmented row. A strict valley would contain such two adjacent differences by telescoping, which proves the claim. □

A.4 A start with one step in each direction

Lemma A.6. *Fix the first two steps to be $+1, -1$. For $0 \leq a \leq 1$, the count law at each length $n \geq 2$ is log-concave, with positive support $\{1, \ldots, n-1\}$. If $1/2 < a < 1$, its limiting density f_a^{+-} is symmetric and log-concave, with $(f_a^{+-})'(t) < 0$ for $t > 0$. Consequently the density conditional on a positive first step satisfies*

$$h_a'(t) > 0 \qquad (t < 0).$$

Proof. Starting with one step in each direction removes the boundary obstruction to the log-concavity induction of [9, Proposition 2.2]. Write one update as $g_k = Af_k + Bf_{k-1}$, where

$$A = \frac{1+a}{2} - \frac{ak}{n}, \qquad B = \frac{1-a}{2} + \frac{a(k-1)}{n}, \qquad d = \frac{a}{n}.$$

For $u = f_{k-2}$, $v = f_{k-1}$, $w = f_k$ and $z = f_{k+1}$, expansion gives

$$\begin{aligned} g_k^2 - g_{k-1}g_{k+1} = (A^2 - d^2)(w^2 - vz) + (B^2 - d^2)(v^2 - uw) \\ + (A - d)(B - d)(wv - uz) + d^2(w - v)^2. \end{aligned}$$

For $2 \leq k \leq n - 1$ we have $A, B \geq d \geq 0$. The preceding row is log-concave with interval support, so all three gaps are nonnegative. At either boundary an adjacent output mass vanishes. Starting from the point mass at one now proves log-concavity by induction. The same recurrence gives strict positivity throughout the support.

Log-linear interpolation of these lattice masses is log-concave. Its integral differs from that of the step interpolation by at most twice the largest atom, which tends to zero because the limiting law has a continuous density. Thus the limit is log-concave. The two-subtree representation gives

$$\mathcal{L}_a^{+-} \overset{d}{=} U^a \mathcal{L}_a' - (1 - U)^a \mathcal{L}_a'',$$

with independent copies and an independent uniform U. This proves symmetry. It also gives positivity and an entire density, by positivity of h_a and the Fourier bound of Lemma A.3. The derivative of $\log f_a^{+-}$ is nonincreasing and vanishes at zero. If it vanished at a positive point, the density would be constant on an interval, contradicting analyticity. Hence $(f_a^{+-})' < 0$ on the positive half-line.

For two positive initial steps, reflection dominance and the first-difference recurrence give nonnegative first differences on the negative half-line. The induction starts from the point mass at count two. Both parents stay in the preceding lower half except at the central difference of an odd row, where reflection dominance gives the required sign. Passing to the derivative limit as in Lemma A.1 gives $(f_a^{++})'(t) \geq 0$ for $t < 0$. Finally $h_a = ((1+a)/2) f_a^{++} + ((1-a)/2) f_a^{+-}$. The second summand has strictly positive derivative there. □

Guérin, Laulin, Raschel and Simon prove positivity of the conditional mode for memory sufficiently close to one [9, Remark 2.1(a)]. The next lemma gives the strict slope at zero for every superdiffusive memory.

Lemma A.7 (the conditional mode is positive)**.** *For every $1/2 < a < 1$, we have $h_a'(0) > 0$. In particular the unique conditional mode lies strictly to the right of zero.*

Proof. We need a lower bound on the reflection difference that stays positive after rescaling and passing to the limit. Write $f_n(k) = \mathbb{P}_+(A_n = k)$ and, on $z \equiv n \pmod 2$, put $d_n(z) = f_n((n+z)/2) - f_n((n-z)/2)$. The reflection induction in Proposition A.5 is strict for $0 < z \leq n$ when the start is positive. Every noncentral upper-half entry has a positive parent with positive coefficient, starting from $d_1(1) = 1$. The recurrence is

$$d_{n+1}(z) = A_n(z) d_n(z+1) + B_n(z) d_n(z-1),$$
$$A_n(z) = \frac{1}{2} - \frac{a(z+1)}{2n}, \qquad B_n(z) = \frac{1}{2} + \frac{a(z-1)}{2n}.$$

Fix $n_0 > 4a$ and divide by z. After multiplication by $\prod_{j=n_0}^{n-1} (1 - 2a/j)^{-1}$, the resulting values $v_n(z)$ satisfy a backward averaging equation with weights

$$q_+ = \frac{A_n(z)(z+1)}{z(1-2a/n)}, \qquad q_- = \frac{B_n(z)(z-1)}{z(1-2a/n)}.$$

The weights sum to one and are nonnegative in the interior band used below. At $z = 1$, the weight of the parent zero vanishes. For the corresponding backward chain,

$$\mathbb{E}[z_n^2 \mid z_{n+1} = z] = \frac{n-4a}{n-2a} z^2 + \frac{3n-4a}{n-2a}.$$

Set $\gamma_n = \Gamma(n-2a)/\Gamma(n-4a) \sim n^{2a}$. Then $Y_n = z_n^2/\gamma_n$ has backward drift $\delta_n = (3n - 4a)/\{(n-2a)\gamma_n\}$, and $\sum_{n \geq n_0} \delta_n \to 0$ as $n_0 \to \infty$.

Fix $R > 0$. Start at $z_N = 1$ for odd N and $z_N = 2$ for even N, and stop at row n_0 or on leaving $0 < z/\sqrt{\gamma_n} \leq R$. For sufficiently large n_0, every band state and its immediate parents lie inside the support. The expected stopped value of Y is at most $z_N^2/\gamma_N + \sum_{n \geq n_0} \delta_n$. At an exit it exceeds R^2, so increasing n_0 and then N makes the exit probability less than one half. All stopped v values are positive. At row n_0 there are only finitely many band states, so their minimum m_0 is positive. The chain reaches that row with probability greater than one half. The bounded stopping argument therefore gives $v_N(z_N) \geq m_0/2$. The product normalizer is asymptotic to a positive constant times N^{2a}, and hence

$$\liminf_N N^{2a} \frac{d_N(z_N)}{z_N} > 0.$$

The first-difference convergence proved in Lemma A.1, applied to the conditional start, identifies the limit as $4h_a'(0)$. Conditional unimodality then puts its mode to the right of zero. □

B An analytic proof of the transition

To prove Theorem 3.2, we turn the sign of the central curvature into a comparison of positive weights. The moment equation of Guérin, Laulin, Raschel and Simon [9, (15)] supplies the integral representation.

Put

$$\chi_k = 4^{-k}\binom{2k}{k}, \qquad H_a(y) = 1 - \sum_{k\geq 1} \frac{\chi_k}{2ak-1} y^k. \tag{B.1}$$

The function H_a decreases strictly on $[0,1)$. For $a = 1$ it is $\sqrt{1-y}$. For $a < 1$ every subtracted coefficient is larger, so $H_a(1) < 0$. The series at one converges, since its coefficients are bounded by a constant depending on a times $\chi_k/(2k-1)$, whose sum is one. There is consequently a unique root $Y_a \in (0,1)$. The coefficients also give

$$H_a(y) - 2ayH_a'(y) = (1-y)^{-1/2}. \tag{B.2}$$

Since $\partial_a H_a(y) > 0$ for $y > 0$ and $H_a'(Y_a) < 0$, the root is smooth and $Y_a' > 0$.

Lemma B.1. *For $1/2 < a < 1$,*

$$g_a''(0) = -\frac{1}{\pi\Gamma(3-3a)} \int_0^{Y_a} y^{1/2} H_a(y)^{2-3a} \big[1 - (1+a)^2 y\big]\, dy. \tag{B.3}$$

The integral is absolutely convergent.

Proof. We multiply the limit by an independent exponential power to cancel the gamma factors in its moments. Let E be an independent exponential random variable of mean one and set $Z_a = E^a \mathcal{L}_a$. With the normalization of [9],

$$m_n = \frac{\Gamma(1+an)}{n!}\mathbb{E}\mathcal{L}_a^n, \qquad m_0 = m_1 = 1.$$

The bound $0 < m_n \leq C_a^n$ follows from [10, Lemma 2.10 and Corollary 2.14]. Applied to the even moments, it gives $\mathbb{E}\exp(\delta|Z_a|) < \infty$ for small $\delta > 0$. Hence $M_a(z) = \mathbb{E}e^{zZ_a}$ is analytic on a strip about the imaginary axis. Its power series and the moment equation give

$$M_a(z) + azM_a'(z) = \frac{1+a}{2} M_a(z)^2 + \frac{1-a}{2} M_a(z) M_a(-z). \tag{B.4}$$

The identity holds on the strip by analytic continuation. If $M_a(it) = A(t) + iB(t)$, it becomes

$$atA' = A^2 - A - aB^2, \qquad atB' = ((1+a)A - 1)B. \tag{B.5}$$

For $0 < y < Y_a$ define

$$t = \sqrt{y}\, H_a(y)^{-a}, \qquad A(t) = H_a(y)\sqrt{1-y}, \qquad B(t) = H_a(y)\sqrt{y}. \tag{B.6}$$

This is the imaginary-axis form of the closed expression in [9, Proposition 3.2], with their $G(it) = i\sqrt{y}$ and $(G(it)/(it))^{1/a} = H_a(y)$. We verify the continuation directly from (B.5). The map from y to t increases from zero to infinity. Equation (B.2) gives

$$\frac{1}{t}\frac{dt}{dy} = \frac{1}{2yH_a(y)\sqrt{1-y}},$$

and substitution shows that (B.6) satisfies (B.5). Since $H_a(0) = 1$, the analytic map $y \mapsto yH_a(y)^{-2a}$ has derivative one at zero. The identity $t^2 = yH_a(y)^{-2a}$ therefore makes y analytic in t^2 near zero. Thus A is even and analytic, while $B = tH_a(y)^{1+a}$ is odd and analytic, with $A(0) = 1$ and $B'(0) = 1$. These data determine both power series. The coefficient recurrence has denominator $na - 1$ for even $n \geq 2$, and $na - a$ for odd $n \geq 3$, both positive. Thus the parametrization agrees locally with $M_a(it)$. Ordinary uniqueness for (B.5) on $t > 0$ extends the agreement to every positive t.

Write $f(t) = \operatorname{Re} \mathbb{E} e^{it\mathcal{L}_a}$. Exponential randomization gives

$$A(t) = \int_0^\infty e^{-v} f(tv^a)\, dv.$$

We use the Fourier decay in Lemma A.3, which in particular makes every polynomial multiple of f integrable at infinity. For $0 < \Re s < 1/a$, absolute Fubini gives

$$J(s) := \int_0^\infty t^{s-1} A(t)\, dt = \Gamma(1 - as)\mathcal{F}(s), \qquad \mathcal{F}(s) = \int_0^\infty t^{s-1} f(t)\, dt. \tag{B.7}$$

The function $\mathcal{F}$ is holomorphic on $\Re s > 0$. The integral defining J diverges at $s = 3$, the value needed for curvature. Two integrations by parts give a convergent expression there.

We send the root to zero before integrating by parts. Set $\theta = \arcsin \sqrt{y}$ and

$$u = F(\theta) = \sin^{-1/a} \theta\, H_a(\sin^2 \theta), \qquad w(u) = a \cos\theta \sin^{1/a} \theta.$$

Here θ decreases from $\arcsin \sqrt{Y_a}$ to zero as u increases from zero to infinity. From (B.2),

$$F'(\theta) = -\frac{1}{a} \sin^{-1-1/a} \theta, \qquad t = u^{-a}.$$

Changing variables in J gives $J(s) = \int_0^\infty u^{-as} w(u)\, du$. Since $F'(\arcsin \sqrt{Y_a}) \neq 0$, the function w is smooth at zero. At infinity, $\theta \sim u^{-a}$, so its definition and the derivative formulas below give $w(u) = O(u^{-1})$, $w'(u) = O(u^{-2})$ and $w''(u) = O(u^{-3})$. For $0 < \Re s < 1/a$, the boundary terms therefore vanish twice, and (B.7) becomes

$$\mathcal{F}(s) = \frac{1}{\Gamma(3 - as)} \int_0^\infty u^{2-as} w''(u)\, du. \tag{B.8}$$

The right side is holomorphic on $0 < \Re s < 3/a$ by the bounds at infinity and boundedness of w'' at zero. Thus (B.8) holds throughout this larger strip by the identity theorem, and $s = 3$ is permitted.

Differentiating through $u = F(\theta)$ gives

$$\begin{aligned} w'(u) &= -a \sin^{2/a} \theta [1 - (1 + a) \sin^2 \theta], \\ w''(u) &= 2a \sin^{3/a} \theta \cos\theta [1 - (1 + a)^2 \sin^2 \theta]. \end{aligned}$$

Insert the second expression in (B.8) at $s = 3$ and return to $y = \sin^2 \theta$. The powers of y reduce to $y^{1/2}$, giving the integral in (B.3). Fourier inversion supplies the factor $g_a''(0) = -\mathcal{F}(3)/\pi$. Finally, H_a has a simple zero at Y_a, and $2 - 3a > -1$, so the endpoint singularity is integrable. □

Proof of Theorem 3.2. Signs outside a short interval.

Let μ_a be the probability law on $(0, Y_a)$ with density proportional to $y^{1/2} H_a(y)^{2-3a}$, and put

$$\mathcal{R}(a) = (1 + a)^2 \int y\, d\mu_a(y). \tag{B.9}$$

Lemma B.1 says that $g_a''(0)$ has the sign of $\mathcal{R}(a) - 1$.

When $a \le 2/3$, the extra weight H_a^{2-3a} decreases in y. By Chebyshev's association inequality, it lowers the mean of the density proportional to $y^{1/2}$, giving $\int y\, d\mu_a(y) \le 3Y_a/5$. At $a = 2/3$, the first three terms of (B.1) give

$$H_{2/3}(3/5) \le 1 - \frac{9}{10} - \frac{81}{1000} - \frac{9}{400} = -\frac{7}{2000} < 0.$$

Since Y_a increases, $Y_a \le Y_{2/3} < 3/5$ for $a \le 2/3$. Thus $\mathcal{R}(a) < (5/3)^2(3/5)^2 = 1$.

When $a \ge 2/3$, the extra weight increases, giving a mean of at least $3Y_a/5$. We need the bound $Y_{2/3} > 117/200$. To prove it, note that χ_k decreases. At $y = 117/200$, the terms with $k \ge 4$ are at most

$$\sum_{k\ge 4} \frac{\chi_k y^k}{4k/3 - 1} \le \frac{\chi_4 y^4}{(13/3)(1-y)}.$$

Together with the first three terms this gives the exact bound

$$H_{2/3}(117/200) \ge \frac{929771}{135987200} > 0.$$

For every $a \ge 69/100$, therefore,

$$\mathcal{R}(a) > \left(\frac{169}{100}\right)^2 \frac{3}{5}\frac{117}{200} = \frac{10024911}{10000000} > 1.$$

Any zero must lie between $2/3$ and $69/100$.

A single crossing.

We now work on $I = [2/3, 69/100]$. To compare densities on a fixed interval, rescale by $y = Y_a u$, so u always lies in $(0, 1)$. We separate the endpoint zero by putting

$$w_k(a) = \frac{\chi_k Y_a^k}{2ak - 1}, \qquad K_a(u) = \frac{H_a(Y_a u)}{1-u} = \sum_{k\ge 1} w_k(a)(1 + u + \cdots + u^{k-1}). \tag{B.10}$$

Since $\sum w_k = 1$, we have $K_a \ge 1$ and $\partial_u K_a \ge 0$. Writing $s_a = 3a - 2 \in [0, 7/100]$, the rescaled measure has density proportional to

$$W_a(u) = u^{1/2}(1-u)^{-s_a} K_a(u)^{-s_a}. \tag{B.11}$$

To make the mean increase, we show that the score $\partial_a \log W_a(u)$ is strictly increasing in u and use the covariance identity for the derivative of an expectation.

First, $Y_a < 2/3$ on I. The first three terms give

$$H_{69/100}(2/3) \le -\frac{9767}{7087608} < 0,$$

and Y_a increases. All series and derivatives used below therefore converge uniformly on $I \times [0, 1]$. Define the falling-factorial moments

$$M_1 = \sum k w_k, \qquad M_2 = \sum k(k-1) w_k, \qquad M_3 = \sum k(k-1)(k-2) w_k.$$

For $G(y) = (1-y)^{-1/2}$, equation (B.2) and its first two derivatives at $Y = Y_a$ give

$$\begin{aligned} 2aM_1 &= G(Y), \\ 2aM_2 &= YG'(Y) - (2a-1)M_1, \\ 2a(M_3 + 2M_2) &= Y^2 G''(Y) + M_2. \end{aligned} \tag{B.12}$$

In particular,

$$0 \le \partial_u K_a(u) \le \frac{M_2}{2} \le \frac{YG'(Y)}{4a} \le \frac{3\sqrt{3}}{8} < \frac{21}{32}. \tag{B.13}$$

We next control memory derivatives of the kernel. Write $d_k = 2k/(2ak-1)$. Differentiating $\sum w_k = 1$ gives

$$\frac{Y_a'}{Y_a} M_1 = \sum d_k w_k \le 6, \qquad \frac{Y_a'}{Y_a} \le \frac{12a}{G(Y)}. \tag{B.14}$$

Here d_k decreases in k and $d_1 \le 6$ on I. The logarithmic derivative $\partial_a \log w_k = kY_a'/Y_a - d_k$ strictly increases in k. Since $1 + u + \cdots + u^{k-1}$ is nondecreasing in k, the covariance identity for derivatives of expectations gives $\partial_a K_a(u) \ge 0$. The probability law here is formed by the w_k.

In $\partial_a \partial_u K_a$, discard the negative term $-d_k w_k$ from $\partial_a w_k$, then bound the derivative of $1 + u + \cdots + u^{k-1}$ by $k(k-1)/2$. Equations (B.12) and (B.14) give

$$\begin{aligned} \partial_a \partial_u K_a &\le \frac{Y_a'}{Y_a} \frac{M_3 + 2M_2}{2} \\ &\le 3\left(\frac{M_2}{G(Y)} + \frac{Y^2 G''(Y)}{G(Y)} \right) \le 3\left(\frac{3}{4} + 3 \right) = \frac{45}{4}. \end{aligned} \tag{B.15}$$

The last bounds use $M_2/G(Y) \le Y/[4a(1-Y)]$ and $Y^2 G''(Y)/G(Y) = 3Y^2/[4(1-Y)^2]$. Since $K_a \ge 1$, $\partial_a K_a \ge 0$ and $\partial_u K_a \ge 0$, we obtain

$$\partial_a \left(\frac{\partial_u K_a}{K_a} \right) = \frac{\partial_a \partial_u K_a}{K_a} - \frac{\partial_u K_a \, \partial_a K_a}{K_a^2} \le \frac{45}{4}.$$

In (B.11), the endpoint term dominates the kernel terms, giving the strict bound

$$\begin{aligned} \partial_u \partial_a \log W_a(u) &= \frac{3}{1-u} - 3\frac{\partial_u K_a}{K_a} - s_a \partial_a \left(\frac{\partial_u K_a}{K_a} \right) \\ &\ge 3 - \frac{63}{32} - \frac{7}{100}\frac{45}{4} = \frac{39}{160} > 0. \end{aligned} \tag{B.16}$$

The score $\partial_a \log W_a$ is therefore strictly increasing in u. If ν_a denotes the normalized measure in (B.11), differentiation gives

$$\frac{d}{da} \int u \, d\nu_a(u) = \mathrm{Cov}_{\nu_a}\big(u, \partial_a \log W_a(u)\big) > 0.$$

This is the strict form of Chebyshev's association inequality. Indeed, the covariance is one half the expectation of $(U-V)(S(U)-S(V))$ for independent $U, V \sim \nu_a$ and the strictly increasing score S. Differentiation under the integral is justified by the integrable bound $C(1-u)^{-7/100}(1 + |\log(1-u)|)$ on this compact parameter interval.

It follows that

$$\mathcal{R}(a) = (1+a)^2 Y_a \int u \, d\nu_a(u)$$

has strictly positive derivative in the interior of I. The signs already proved and the intermediate value theorem give exactly one zero of $\mathcal{R} - 1$. There are no zeros outside I. Finally, (B.3) writes

$$g_a''(0) = \frac{\mathcal{Z}_a}{\pi \Gamma(3-3a)} \, [\mathcal{R}(a) - 1], \qquad \mathcal{Z}_a = \int_0^{Y_a} y^{1/2} H_a(y)^{2-3a} \, dy > 0.$$

The normalizing factor is continuously differentiable near the zero, by the same rescaling and domination. At the zero its derivative contributes nothing, so $\partial_a g_a''(0) =$

$\mathcal{Z}_a\mathcal{R}'(a)/[\pi\Gamma(3-3a)] > 0$. The shape alternatives in Theorem 3.1 now give unimodality below a_* and bimodality above it. At the critical point, let $a \uparrow a_*$ from below. Joint continuity preserves $g_a(x) \geq g_a(y)$ for $0 \leq x < y$. Thus the critical density is nonincreasing on the positive half-line too. Analyticity excludes a constant interval, so its unique mode is at zero. □

The root bounds also determine the sign of the fourth derivative at the threshold.

Lemma B.2. *For $2/3 \leq a \leq 69/100$, we have $g_a^{(4)}(0) < 0$. In particular, $B_* = -g_{a_*}^{(4)}(0) > 0$.*

Proof. We first extend the Mellin calculation by one integration by parts. With u, w as in the proof of Lemma B.1, put

$$c_a = (1+a)^2, \qquad P_a(y) = \frac{3}{a}(1-y)(1-c_a y) - y(1-c_a y) - 2c_a y(1-y).$$

Since $dy/du = -2ay^{1+1/(2a)}\sqrt{1-y}$, differentiation gives $w'''(u) = -2a^2 y^{2/a} P_a(y)$. This derivative is bounded at zero and is $O(u^{-4})$ at infinity. The boundary terms vanish for $0 < \Re s < 3/a$, so (B.8) gives

$$\mathcal{F}(s) = -\frac{1}{\Gamma(4-as)} \int_0^\infty u^{3-as} w'''(u)\, du.$$

The right side is holomorphic for $0 < \Re s < 4/a$. For $1/2 < a < 4/5$, we may therefore set $s = 5$ and use $g_a^{(4)}(0) = \mathcal{F}(5)/\pi$ to obtain

$$g_a^{(4)}(0) = \frac{a}{\pi\Gamma(4-5a)} \int_0^{Y_a} y^{3/2}(1-y)^{-1/2} H_a(y)^{3-5a} P_a(y)\, dy. \tag{B.17}$$

The integral converges absolutely because H_a has a simple zero and $3 - 5a > -1$.

Now let $a \in [2/3, 69/100]$. The root bounds in the preceding proof give $1/2 < Y_a < 2/3$. For the coefficient comparison only, we use the larger interval $2/3 \leq a \leq 7/10$. Expanding P_a gives, for $y \geq 0$,

$$\begin{aligned} P_a(y) &= \frac{3}{a} - \left(2a^2 + 7a + 9 + \frac{6}{a}\right) y + \left(3a^2 + 9a + 9 + \frac{3}{a}\right) y^2 \\ &\leq \frac{9}{2} - 23y + 22y^2 =: Q(y). \end{aligned}$$

Here termwise lower bounds give $8/9 + 14/3 + 9 + 60/7 = 1457/63 > 23$ for the coefficient of $-y$, and termwise upper bounds give $147/100 + 63/10 + 9 + 9/2 = 2127/100 < 22$ for that of y^2. Since $Q(0) > 0$, $Q(1/2) = -3/2$ and $Q(2/3) = -19/18$, this quadratic has exactly one root r in $(0, Y_a)$. It is positive before r and negative after it. For $1/2 \leq Y \leq 2/3$, direct integration gives

$$Y^{-5/2} \int_0^Y y^{3/2} Q(y)\, dy = \frac{9}{5} - \frac{46}{7} Y + \frac{44}{9} Y^2 \leq -\frac{83}{315} < 0,$$

where convexity reduces the bound to the two endpoints.

The remaining weight $R_a(y) = (1-y)^{-1/2} H_a(y)^{3-5a}$ is positive and increasing on $(0, Y_a)$. The sign change at r therefore gives $R_a(y)Q(y) \leq R_a(r)Q(y)$ throughout this interval. Consequently,

$$\int_0^{Y_a} y^{3/2} R_a(y) P_a(y)\, dy \leq R_a(r) \int_0^{Y_a} y^{3/2} Q(y)\, dy < 0.$$

The prefactor in (B.17) is positive on $[2/3, 69/100]$, which proves the claim. □

C Numerical enclosures

The analytic proofs determine the transition and the critical exponents. The computations below only locate the threshold and evaluate their constants.

Corollary C.1 (a certified location)**.** *The unique threshold in Theorem 3.2 satisfies* $z_- < a_* < z_+$*, where*

$$z_- = 1838418790980431/2748779069440000 = 0.668812860014597754\ldots,$$
$$z_+ = 45960469774511/68719476736000 = 0.668812860014601028\ldots.$$

The width of this interval is less than $3.3 \cdot 10^{-15}$*.*

Proof. The endpoint evaluations give $g''_{z_-}(0) < 0 < g''_{z_+}(0)$. Theorem 3.2 identifies the zero between them. Supplementary Sections S2 and S2.3 prove the moment, truncation and Fourier-tail bounds used in these interval evaluations. The endpoint calculation is independent of parameter Taylor models, while both checks use Arb and the same analytic tail bound. □

Constant	Lower bound	Upper bound
C_*	0.6889503945	0.6889503947
B_*	0.079207784468	0.079207784470
H_*	0.09707494510086	0.09707494510088
Q_*	9.86698768	9.86698770
$\sqrt{6C_*/B_*}$	7.224131934	7.224131936

Table 1: The constants in the critical expansions. The enclosure is uniform on a parameter interval containing the certified bracket for a_* (Supplementary Section S2.3).

D Signs of the second difference

We prove the sign bounds of Section 2.

D.1 Averaging adjacent entries

Insert each η_i between its two neighbors in ζ. If $\eta_i = 0$, the sign pattern does not change. If $\eta_i > 0$, then $\zeta_i > 0$ or $\zeta_{i+1} > 0$, so η_i is placed next to an entry of its own sign, and the pattern does not change. The same holds for $\eta_i < 0$, proving the first claim. For the second, obtain η from w by deleting the entries ζ_i. Deletion can only remove signs from the pattern. □

D.2 Signs at the first lattice point

Put $\xi_t = P_t(1)/P_t(0)$. By (2.1), $P_{t+1}(0) = (1-\varepsilon)P_t(0)$ and $P_{t+1}(1) = (1-\varepsilon-\frac{a}{t})P_t(1) + \varepsilon P_t(0)$. So

$$\xi_{t+1} - \xi_t = \frac{1}{1-\varepsilon}\Big(\varepsilon - \frac{a\xi_t}{t}\Big), \qquad \xi_2 = \frac{2\varepsilon}{1-\varepsilon}.$$

We claim that $\xi_t < \varepsilon t/a$ for $t \geq 2$. For $t = 2$ this holds because $a < 1-\varepsilon$. If it holds for t, then $0 < \xi_{t+1} - \xi_t \leq \frac{\varepsilon}{1-\varepsilon} < \frac{\varepsilon}{a}$, and it holds for $t+1$. So ξ_t is strictly increasing for $t \geq 2$, and $\xi_t < 1$ whenever $t \leq a/\varepsilon$.

Now (2.2) with $k = 1$, and $D_t(0) = P_t(1) - 2P_t(0)$, give for $t \geq 2$

$$s_{t+1} = \big(1 - r_t(2)\big)s_t + \sigma_t, \qquad \sigma_t = \Big(\varepsilon - \frac{a}{t}\Big)(\xi_t - 2)\, P_t(0).$$

Here $1 - r_t(2) \geq 1 - \varepsilon - a = \varepsilon > 0$. If $t < a/\varepsilon$, both factors of σ_t are negative and $\sigma_t > 0$. If $t > a/\varepsilon$, the first factor is positive, and the second is first negative and then

positive, because ξ_t increases. So $\mathrm{sp}(\sigma_2, \sigma_3, \dots) \preceq \Pi_3$. To remove the positive multiplier, let $\Phi_t = \prod_{u=2}^{t-1}(1 - r_u(2)) > 0$. Then

$$\frac{s_t}{\Phi_t} = s_2 + \sum_{u=2}^{t-1} \frac{\sigma_u}{\Phi_{u+1}}, \qquad s_2 = 1 - 3\varepsilon.$$

If $\varepsilon < \frac{1}{3}$, the terms $s_2, \sigma_2, \sigma_3, \dots$ are first nonnegative, then nonpositive, then nonnegative. Their partial sums are nonnegative and nondecreasing in the first stretch, nonincreasing in the second, and nondecreasing in the third. So the sign pattern of the partial sums is a subsequence of Π_3. If $\varepsilon \geq \frac{1}{3}$, then $a/\varepsilon \leq 1$, so the terms are first nonpositive and then nonnegative, and the sign pattern of the partial sums is a subsequence of $(-, +)$. This proves the claim.

□

D.3 At most five sign blocks

For $3 \leq t \leq N$, we retain the endpoint history in the list $\mathcal{Z}_t$

$$\mathcal{Z}_t = \big(s_N, s_{N-1}, \dots, s_{t+1},\ s_t, D_t(2), D_t(3), \dots, D_t(t-2), s_t,\ s_{t+1}, \dots, s_{N-1}, s_N\big).$$

It has $2N-t-1$ entries. Its middle part is $(D_t(1), D_t(2), \dots, D_t(t-1))$, because $D_t(t-1) = D_t(1) = s_t$ by symmetry. The list $\mathcal{Z}_3$ is $(s_N, \dots, s_4, s_3, s_3, s_4, \dots, s_N)$, and $\mathcal{Z}_N$ is the list in the statement.

To apply Lemma 2.3, we check how $\mathcal{Z}_{t+1}$ is obtained from $\mathcal{Z}_t$ for $3 \leq t \leq N-1$. The first $N-t$ entries of $\mathcal{Z}_{t+1}$ are $s_N, \dots, s_{t+1}$, and they are copies of the first $N-t$ entries of $\mathcal{Z}_t$. So $A_i = 1$ and $B_i = 0$ there. The next $t-2$ entries are $D_{t+1}(b)$ for $2 \leq b \leq t-1$. By (2.2),

$$D_{t+1}(b) = r_t(b-2)\, D_t(b-1) + \big(1 - r_t(b+1)\big) D_t(b),$$

and $D_t(b-1)$ and $D_t(b)$ are neighbors in the middle part of $\mathcal{Z}_t$, in the required order. The coefficients satisfy $r_t(b-2) \geq \varepsilon$ and $1 - r_t(b+1) \geq 1 - \varepsilon - a = \varepsilon$, since $2 \leq b \leq t-1$. The last $N-t$ entries of $\mathcal{Z}_{t+1}$ are $s_{t+1}, \dots, s_N$, and each is a copy of the entry of $\mathcal{Z}_t$ one place to its right. So $A_i = 0$ and $B_i = 1$ there.

By Lemma 2.3, $\mathrm{sp}(\mathcal{Z}_N) \preceq \mathrm{sp}(\mathcal{Z}_{N-1}) \preceq \cdots \preceq \mathrm{sp}(\mathcal{Z}_3)$. By Lemma 2.4 the sign pattern p of $(s_3, \dots, s_N)$ is a subsequence of Π_3, so it is one of the lists $()$, $(+)$, $(-)$, $(+,-)$, $(-,+)$ and $(+,-,+)$. The sign pattern of $\mathcal{Z}_3$ is p reversed, followed by p, with equal neighbors merged. In the six cases this is $()$, $(+)$, $(-)$, $(-,+,-)$, $(+,-,+)$ and $(+,-,+,-,+)$, and each is a subsequence of Π_5.

□

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